

The total number of points for each problem is indicated.

Each part of a problem is worth the same amount.

1. (16 points) Give an example of each of the following, or explain why it is not possible.

a) Two matrices that are similar but not row equivalent.

b) A finite-dimensional vector space V and an operator A such that A has infinitely many eigenvectors.

c) An operator A on $\mathbb{C}^5$ such that its column space and nullspace are equal.

d) A matrix that satisfies the polynomial equation $x^5 = 0$ but that does not satisfy $x^4 = 0$.

e) Two matrices with equal characteristic polynomials and different determinants.

f) An operator A on a vector space V such that there is an A -invariant subspace of V that is not an eigenspace.

g) A square matrix A such that $A^2 = A$ and so that A is neither an identity matrix nor a zero matrix

h) A square matrix that has an LU decomposition but does not have a QR decomposition.

2. (6 points) Give a precise definition for each of the following linear algebra terms.

similar:

eigenvector:

3. (9 points) Complete the following:

a) State the Spectral Theorem. Give an example that illustrates it.

b) Give three examples of conditions that are equivalent to the following: the columns of an $n \times n$ matrix A are linearly *dependent*.

c) In some combination, clearly state three ways in which the trace, determinant, eigenvalues, singular values, and characteristic polynomial of a square matrix are related.

4. (12 points) Prove two of the following:

- a) The linear transformation represented by a square matrix A is onto if and only if the linear transformation represented by the square matrix A^3 is onto.
- b) If v_1 , v_2 , and v_3 are linearly independent, then $v_1 + v_2$, $v_2 + v_3$, and $v_3 + v_1$ are linearly independent.
- c) An $n \times n$ complex nilpotent matrix A satisfies the equation $x^n = 0$, where 0 is the zero matrix.
- d) If A is a real square matrix, then A and A^T have the same set of eigenvalues, each with the same algebraic multiplicity.

5. (15 points) Always, sometimes, or never? If it is always or never true, give a proof. If it is sometimes true, give an example where it's true and where it's false.

a) A product of an orthogonal matrix and a nilpotent matrix is orthogonal.

b) A real 2×2 matrix is matrix equivalent to either $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ or $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

c) A square matrix over $\mathbb{C}$ is similar to an upper triangular matrix.

d) A homogeneous system of linear equations over $\mathbb{R}$ has infinitely many solutions.

e) The characteristic polynomial and the minimal polynomial of a 3×3 real matrix are equal.

6. (4 points)

Define the following function q from the vector space of 2×2 real matrices to $\mathbb{R}$.

$$q : \begin{pmatrix} a & c \\ b & d \end{pmatrix} \mapsto ad + bc$$

a) Is q a determinant function? Explain why q does or does not satisfy the properties of a determinant.

b) Is q a norm on this vector space? Explain why q does or does not satisfy the properties of a norm.

7. (12 points)

a) Circle two matrices that are unitary.

$$\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \quad \begin{pmatrix} i & 1 \\ -1 & -i \end{pmatrix} \quad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \begin{pmatrix} -1 & 1+2i \\ 1-2i & 0 \end{pmatrix} \quad \begin{pmatrix} i & 0 \\ 0 & 0 \end{pmatrix}$$

b) Circle two matrices that are diagonalizable over $\mathbb{R}$.

$$\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \quad \begin{pmatrix} 4 & 0 \\ 0 & 0 \end{pmatrix} \quad \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad \begin{pmatrix} 1 & 4 \\ 0 & 1 \end{pmatrix} \quad \begin{pmatrix} 4 & 1 \\ 0 & 1 \end{pmatrix} \quad \begin{pmatrix} 0 & 5 \\ 0 & 0 \end{pmatrix}$$

c) Circle two matrices that are in reduced row echelon form.

$$\begin{pmatrix} 1 & 7 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \begin{pmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \begin{pmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix} \quad \begin{pmatrix} 0 & 4 \\ 0 & 1 \end{pmatrix} \quad \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

d) Circle two types of matrices that form subspaces of 2×2 real matrices.

nilpotent invertible symmetric orthogonal upper triangular identity

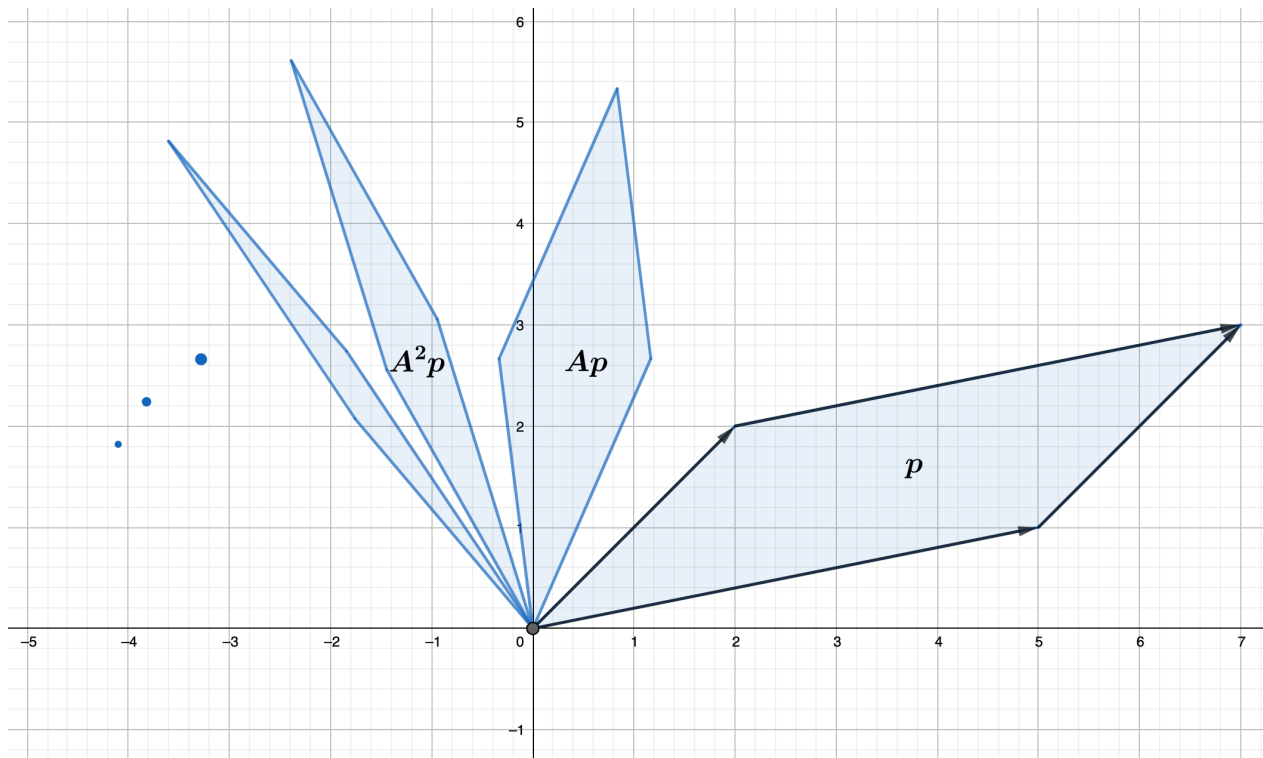
e) Circle two inner products on $\mathbb{R}^2$.

$$\langle x, y \rangle = x_1 y_1 \quad \langle x, y \rangle = x_2 y_2 + x_1 y_1 \quad \langle x, y \rangle = 0 \quad \langle x, y \rangle = 5x_1 y_1 + 5x_2 y_2 \quad \langle x, y \rangle = \|x\| + \|y\|$$

f) If A has -1 as an eigenvalue, circle two matrices that must also have -1 as an eigenvalue.

$$A^2 \quad A^5 \quad A^2 - 2I \quad -A \quad A^{-1} \quad A^2 + A$$

8. (6 points)



- Let p be the parallelogram defined by the vectors $\begin{pmatrix} 5 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 2 \end{pmatrix}$, as illustrated in the diagram above. Compute the area of p .
- Let A be the linear operator $\begin{pmatrix} 1/3 & -1/2 \\ 1/3 & 1 \end{pmatrix}$, written using the standard basis, and let Ap be the image of p under A . Find the area of Ap .
- Consider the infinite sequence of parallelograms $p, Ap, A^2p, A^3p, \dots$, and take the sum of their areas, considered as an infinite series. If this series converges, compute the sum. If it diverges, explain why.

9 & 10. (20 points)

Complete the following. Use the next three blank pages, as needed. Label each part of your solution clearly, (a) through (j). (These need not be completed in order.)

$$\text{Let } A = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 6 & 3 \\ 0 & -4 & -1 \end{pmatrix}$$

- a) Compute the rank and determinant of A .
- b) Compute the characteristic polynomial and eigenvalues of A .
- c) Compute bases for the eigenspaces of A .
- d) Give the general solution to the matrix equation $Ax = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$.
- e) Compute the result of applying A^{50} to the vector $\begin{pmatrix} 0 \\ -3 \\ 4 \end{pmatrix}$.
- f) Compute the singular values of A , or explain why this is not possible.
- g) Find a real matrix B such that $B^2 = A$, or explain why no such B exists.
- h) Give the Jordan canonical form of A ; or explain why A cannot be put in this form.
- i) Compute a basis for the orthogonal complement of the column space of A .
- j) Circle the terms that apply to A , and do not circle the ones that don't:
nilpotent invertible unitary triangular diagonalizable permutation self-adjoint

See the last page of the packet for the bonus problem.

Bonus: Explain one or more linear algebra ideas or results that you learned in the course but that haven't been asked about on this exam.